MTH102

Linear Algebra

2025–2026–I Semester

03 September 2025

Chapter 1. Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** during **5:00pm to 6:00pm** on

- Mondays,
- Tuesdays,
- Wednesdays,
- Fridays.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

Calendar

The academic calendar is available at https://www.iiserb.ac.in/doaa/schedule.

The announcements related to the course will be made on the following Google classroom page.

https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0

You could also access the notes at the link below (until you have joined the Google classroom).

https://jpsaha.github.io/home/blog/2025/mth102/

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Introduction

Chapter 2. Sets

Definition

A **set** is a well-defined collection of objects. The objects of a set are called its **elements** or **members**.

Examples

- The set of vowels a, e, i, o, u.
- The set consisting of 2, 4, 6, 8, 10.
- The set consisting of 3, 15, 35, 63, 99.

Here are further examples.

Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.

Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.

Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A,
- b to denote an element of B,
- c to denote an element of C,
- x to denote an element of X,
- y to denote an element of Y,
- z to denote an element of Z.

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
$\mathbb{Z}$	the set of integers

If x is an element of a set X, then one says that x **belongs to X**, and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{\text{a, e, i, o, u}\}.$$

Note that the elements are seperated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even}, x > 0\}.$$

This reads

"E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$".

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since $1, 2$ are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \ldots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{\frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0\right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**.

Example

The sets

$$\{\text{a, e, i, o, u}\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 2.1. Determine the elements of the following sets.
- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

§2.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B, then we say that A is a **subset** of B, and write

$$A \subseteq B.$$

If A is not a subset of B, one writes

$$A \nsubseteq B.$$

Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes

$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$

- Also note that

$$\mathbb{Z} \nsubseteq \mathbb{N}, \mathbb{Q} \nsubseteq \mathbb{Z}, \mathbb{Q} \nsubseteq \mathbb{N}.$$

If A is a subset of B and A is not equal to B, then A is said to be a *proper subset of* B, and one writes

$$A \subsetneq B.$$

Note that

$$\mathbb{N} \subsetneq \mathbb{Z}, \mathbb{Z} \subsetneq \mathbb{Q}.$$

Exercise 2.2. What does it mean to say that A is not a subset of B?

Exercise 2.3. Show that if A is a subset of B and B is a subset of C, then A is a subset of C.

Exercise 2.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Consider the set

$$A = \{n \in \mathbb{N} : n^4 - 5n^2 + 6 = 0\}.$$

Note that the set A has no elements. Any such set is called the **empty set** or **null set**, and is denoted by the symbol

$$\emptyset.$$

Two sets, A and B, are said to be **equal**[1] if any element of A belongs to B, and any element of B also belongs to A, or equivalently,

$$A \subseteq B \text{ and } B \subseteq A$$

holds. If two sets A, B are not equal, one writes

$$A \neq B.$$

Exercise 2.5. What does it mean to say that two sets A, B are not equal?

§2.2 Operations on sets

Exercise 2.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

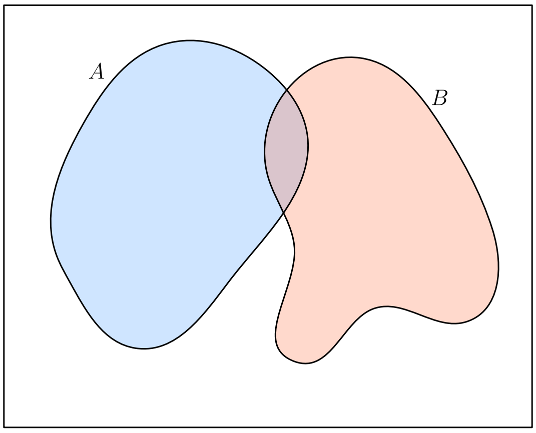

Figure 1: Two sets A and B

Definition

If A, B are sets, their **union** is denoted by $A \cup B$, which is defined as

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

See Figure 2.

[1]*It* may appear obvious! The advantage of putting forth definitions is to set up notations and conventions, so that these are not left to interpretations!

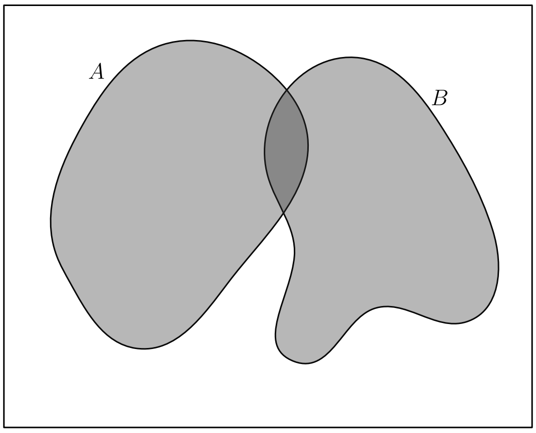

Figure 2: The union of A and B

Exercise 2.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Usage of "or" in the inclusive sense

The word "or" is used in the inclusive sense, allowing two or more of the conditions to be satisfied. In common terminology, this inclusive sense is denoted by "and/or".

Exercise 2.8. If A is a set, show that

$$A \cup A = A.$$

Exercise 2.9 (Commutative property)**.** If A, B are sets, show that

$$A \cup B = B \cup A.$$

Definition

If A, B, C are sets, their **union** is denoted by $A \cup B \cup C$, which is defined as

$$A \cup B \cup C = \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

Exercise 2.10 (Associative property)**.** If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Use Exercise 2.9 and Exercise 2.10, to show that for any three sets A, B, C, the following sets are equal

$$A \cup B \cup C, A \cup C \cup B, B \cup A \cup C, B \cup C \cup A, C \cup A \cup B, C \cup B \cup A.$$

Definition

If A, B are sets, their **intersection** is denoted by $A \cap B$, which is defined as

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

See Figure 3.

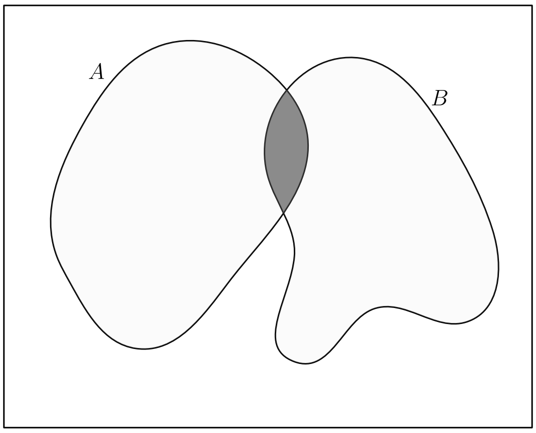

Figure 3: The intersection of A and B

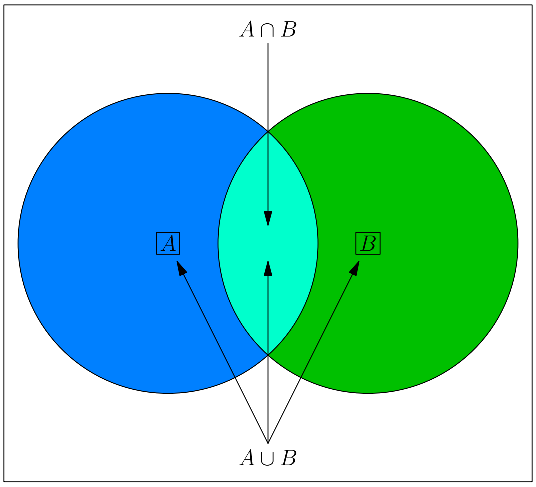

Figure 4: The union and intersection of A and B

Lemma 2.11. *Let X be a set, and A, B be finite subsets of X. Then*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Exercise 2.12. Using Lemma 2.11 or otherwise, show that for finite subsets A, B, C of a set X,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Exercise 2.13. If A is a set, show that

$$A \cap A = A.$$

Exercise 2.14 (Commutative property)**.** If A, B are sets, show that

$$A \cap B = B \cap A.$$

Definition

If A, B, C are sets, their **intersection** is denoted by $A \cap B \cap C$, which is defined as

$$A \cap B \cap C := \{x : x \in A \text{ and } x \in B \text{ and } x \in C\}.$$

Exercise 2.15 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 2.16 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Remark

All sets under consideration in a given context, are assumed to be contained in a 'large' fixed set, called the *universal set*. For example, while studying the positive integers (for instance, some sets consisting of the positive integers), the set $\mathbb{N}$ can be taken as the universal set. While studying the integers, the set $\mathbb{Z}$ can be taken as the universal set.

Definition

Let A, B be subsets of X. The **complement of A in B** is denoted by $B \setminus A$, and is defined by

$$B \setminus A := \{x \in X : x \in B, x \notin A\}$$

It is also called the **difference of B and A**.

The complement of A in X, is called the *complement of* A, and is denoted by A^c, and is defined to be

$$A^c := \{x \in X : x \notin A\}$$

Exercise 2.17. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 2.18. If A, B are subsets of X, show that

$$A^c \cap B = B \setminus A.$$

Exercise 2.19. If A is a subset of X, then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 2.20. If A, B are subsets of X, then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Definition

Two sets A, B are said to be **disjoint** if

$$A \cap B = \emptyset.$$

Exercise 2.21. Show that for any sets A, B, the following inclusions hold.

$$\begin{aligned} A \cap B &\subseteq A, \\ A \cap B &\subseteq B, \\ A &\subseteq A \cup B, \\ B &\subseteq A \cup B. \end{aligned}$$

Exercise 2.22. Show that for any sets A, B, the following are equivalent.

$$\begin{gathered} A \subseteq B, \\ A \cap B = A, \\ A \cup B = B, \\ B^c \subseteq A^c. \end{gathered}$$

Exercise 2.23. If A, B are sets, show that

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$,
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Definition

The **symmetric difference** of two sets A, B, denoted by $A \Delta B$, is defined as

$$A \Delta B = (A \cup B) \setminus (A \cap B).$$

Exercise 2.24. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 2.25. If A, B are sets, then show that

$$\begin{gathered} A \Delta B = B \Delta A, \\ A \cup B = (A \Delta B) \cup (A \cap B), \end{gathered}$$

and

$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

Exercise 2.26. Do there exist three finite sets A, B, C satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$